Supplementary Material

Supplementary Material for "Few-Shot Specific Emitter Identification: A Signal-Type Mismatched Learning Approach"

I. Sensitivity Analysis of Key Hyperparameters

We report the sensitivity analyses of the key hyperparameters of the proposed method, including the masking ratio ρ, the feature distribution perturbation (FDP) coefficient λ, the FDP activation probability p, the aggregation depth L, and the self-distillation strategy. All analyses are conducted on the WiFi dataset under the 20-shot signal-type mismatched setting (MS), i.e., the model is pre-trained on ADS-B and fine-tuned on WiFi, and each accuracy value is averaged over the 50 random seeds. The results are summarized in Figs. S1–S5.

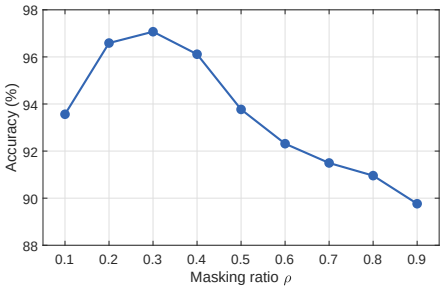


Fig. S1: Sensitivity of masking ratio ρ (WiFi, 20-shot, MS).

A. Masking Ratio ρ

On the WiFi dataset (20-shot, MS), the masking ratio is varied from 0.1 to 0.9 with a step of 0.1. The identification accuracy first increases and then decreases, reaching the maximum at $\rho = 0.3$ (97.07%), while the accuracy is 93.56% at $\rho = 0.1$ and drops to 89.76% at $\rho = 0.9$ as the masking becomes too aggressive. A ratio that is too small leaves the pretext task trivial, whereas one that is too large removes so much context that the latent target becomes unrecoverable, so the optimum at $\rho = 0.3$ therefore reflects a balance between pretext-task difficulty and retained context. The results are shown in Fig. S1.

B. Perturbation Coefficient λ

On the WiFi dataset (20-shot, MS), the coefficient λ is varied from 0.01 to 0.9, with a step of 0.01 below 0.1 and a step of 0.1 above, and the working point is $\lambda = 0.05$. The accuracy peaks exactly at the working point (97.07%), with 96.13% at $\lambda = 0.04$, and then decreases as λ increases, i.e., as the perturbation weakens, dropping to 87.23% at $\lambda = 0.9$. Because λ weights the instance's own statistics against those of a shuffled instance, a small λ (i.e., strong perturbation) prevents the encoder from relying on auxiliary-data-specific statistics, which is precisely the effect the FDP is designed for. The results are shown in Fig. S2.

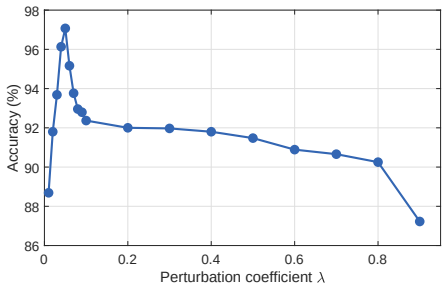


Fig. S2: Sensitivity of perturbation coefficient λ (WiFi, 20-shot, MS).

C. FDP Activation Probability p

On the WiFi dataset (20-shot, MS), the activation probability is varied from 0.1 to 0.9 with a step of 0.1 and is optimal at $p = 0.5$ (97.07%), with the accuracy staying within 92.11–97.07% over the whole range. Activating the perturbation on roughly half of the forward passes is the most favourable setting: a lower p perturbs the auxiliary-specific statistics insufficiently, whereas a higher p discards too much instance-specific information. The results are shown in Fig. S3.

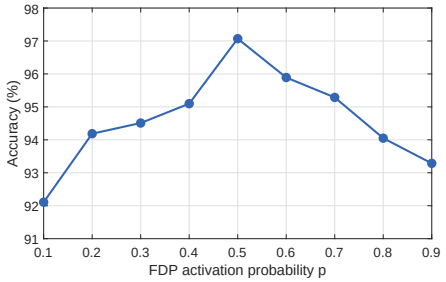


Fig. S3: Sensitivity of FDP activation probability p (WiFi, 20-shot, MS).

D. Aggregation Depth L

On the WiFi dataset (20-shot, MS), the aggregation depth is varied from 1 to 9 layers. The accuracy increases overall with L, from 90.24% at $L = 1$ to 97.07% at $L = 9$, and it reaches 96.91% at $L = 7$, and aggregating all nine layers gives the best result, confirming the benefit of multi-layer aggregation. Most of the gain is obtained once the shallow layers are included ($L = 1 \rightarrow 7$), suggesting that early-layer features carry complementary RFF information that the deepest layer alone does not encode, after which the gains saturate. The results are shown in Fig. S4.

E. Self-Distillation Strategy

On the WiFi dataset (20-shot, MS), the EMA-teacher variant (the proposed design) achieves 97.07±1.47%, outperforming the stop-gradient-only variant (96.23±1.92%) and the variant without distillation (93.82±2.53%). This indicates that the EMA teacher supplies a more stable target than its stop-gradient counterpart, and that the warm-up stage itself, rather than the EMA update alone, accounts for much of the gain. The results are shown in Fig. S5.

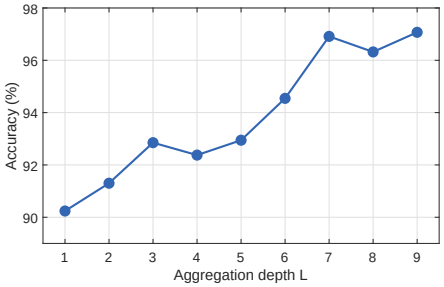


Fig. S4: Sensitivity of aggregation depth L (WiFi, 20-shot, MS).

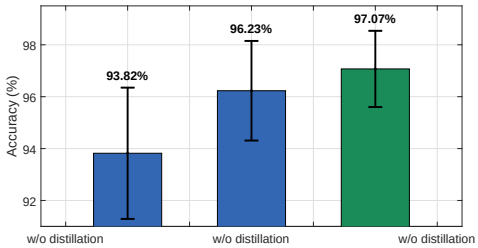


Fig. S5: Self-distillation strategy ablation (WiFi, 20-shot, MS).

II. Reversed Pre-training/Downstream Combinations

To demonstrate the general applicability beyond the ADS-B-based auxiliary pre-training, we additionally conduct experiments with reversed pre-training/downstream combinations, i.e., LoRa $\rightarrow$ WiFi, WiFi $\rightarrow$ LoRa, LoRa $\rightarrow$ ADS-B, and WiFi $\rightarrow$ ADS-B, under the 20-shot setting. The results are summarized in Table S1, which show that the proposed method consistently outperforms AMAE under all reversed combinations (by 1.43–6.47% in absolute accuracy). The absolute accuracy differs across directions because the downstream datasets differ in size (10 categories for ADS-B, 20 for LoRa, and 16 for WiFi) and in signal characteristics, whereas the margin over AMAE remains positive in all four directions.

TABLE S1: Performance under reversed pre-training/downstream combinations (20-shot).

Methods	LoRa$\rightarrow$WiFi	WiFi$\rightarrow$LoRa	LoRa$\rightarrow$ADS-B	WiFi$\rightarrow$ADS-B
Proposed	96.83±1.77%	90.13±3.16%	82.41±2.53%	83.29±1.94%
AMAE	90.36±1.58%	85.37±2.31%	80.98±2.17%	81.35±1.66%

III. Robustness to SNR, Channel Fading and Frequency Offset

All three datasets are open-source recordings captured over the air. The ADS-B recordings contain six emitters captured in an open scenario over four days at 1090 MHz with $L = 256$ samples. The LoRa recordings contain 60 commercial off-the-shelf devices of four models received by a USRP N210 under line-of-sight, non-line-of-sight, static and mobile conditions, and the WiFi recordings contain 16 USRP X310 transmitters together with a set of commercial devices received by a USRP B210 in an open low-reflection area. Because the recordings are made over the air, additive noise, channel fading and carrier frequency offset are physically present in every sample, and the dataset providers report the same for the LoRa and WiFi recordings: in the LoRa recordings the multipath and Doppler effects are explicitly observed under LOS/NLOS and mobile conditions, and the channel model of [6] uses an RMS delay spread up to 300 ns and a Doppler shift up to 10 Hz, while in the ORACLE recordings the transmitter–receiver separation is varied so that the received power changes accordingly. Since these impairments are already present in the recorded samples, we do not replace them but add one controlled layer of a single impairment on top of them, and keep the others at their original level. The experiments are carried out on the WiFi test set (20-shot, MS), and for AWGN and fading the injected level is identical in training and test.

(a) SNR. The noise is scaled to each packet's power,

$$\begin{aligned} y_c[n] &= s_c[n] + w_c[n], \qquad w_c[n] \sim \mathcal{N}\big(0, \sigma_w^2\big)\,, \\ \sigma_w^2 &= \frac{\bar{P}}{10^{\mathrm{SNR}/10}}, \qquad \bar{P} = \frac{1}{2L}\sum_{c}\sum_{n=1}^{L} s_c^2[n], \end{aligned} \tag{1}$$

where $c \in \{\mathrm{I}, \mathrm{Q}\}$ and $s_c[n]$ is the recorded signal.

(b) Channel fading. A four-tap Rician channel with an exponential power-delay profile $p_i \propto e^{-i/2}$ is applied to each packet,

$$y[n] = \sum_{i=0}^{3} h_i\, s[n - d_i], \qquad d_i = 8i, \tag{2}$$

$$h_i = \underbrace{\sqrt{\tfrac{K}{K+1} p_i}\, e^{j\theta_i}\, \delta_{i0}}_{\text{LOS}} + \underbrace{\sqrt{\tfrac{1}{K+1} p_i}\, (a_i + j b_i)}_{\text{scatter}}, \tag{3}$$

where $a_i, b_i \sim \mathcal{N}(0, \frac{1}{2})$ and $\theta_i \sim \mathcal{U}[0, 2\pi)$.

The accuracy versus the impairment level is given in Figs. S6 and S7.

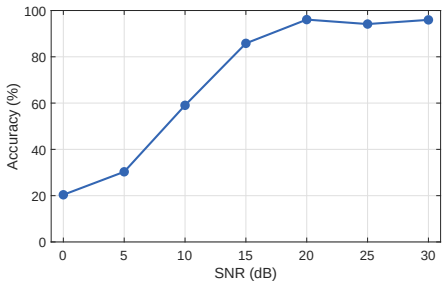


Fig. S6: Identification accuracy under different SNRs (WiFi, 20-shot, MS).

(c) Carrier frequency offset. A residual offset ε between the transmitter and the receiver is applied to each packet,

$$\begin{aligned} y_c[n] &= \Re\Big\{ \big(s_{\mathrm{I}}[n] + j s_{\mathrm{Q}}[n]\big)\, e^{j 2\pi \Delta f\, n / f_{\mathrm{s}}} \Big\}\,, \\ \Delta f &= \varepsilon \cdot 10^{-6} f_{\mathrm{rf}}, \end{aligned} \tag{4}$$

where f_{s} is the sampling rate and f_{rf} the carrier frequency. Fig. S8 reports the accuracy against the offset applied at test time.

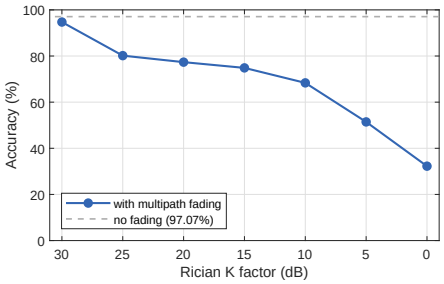


Fig. S7: Identification accuracy versus the Rician K factor of a four-tap multipath channel (WiFi, 20-shot, MS). The dashed line is the fade-free accuracy.

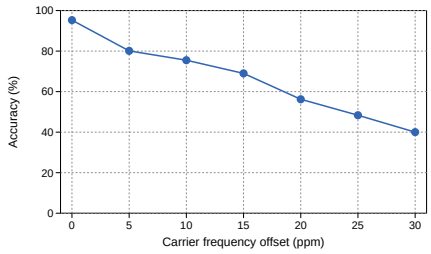


Fig. S8: Identification accuracy versus carrier frequency offset (WiFi, 20-shot, MS).

IV. Quantitative Feature-Space Analysis

Corresponding to the feature visualization (now reported in the main text Fig. 3), we report the Silhouette coefficient (SC), the normalized mutual information (NMI), and the adjusted mutual information (AMI) of the learned features under the 20-shot setting in Table S2. The quantitative metrics are consistent with the visualization: on both datasets the features form well-separated and compact clusters, with WiFi exhibiting the higher values, which is consistent with the larger inter-class separation visible in Fig. 3.

TABLE S2: Quantitative feature-space analysis under the 20-shot setting.

Metrics	LoRa	WiFi
SC (↑)	0.5605	0.6261
NMI (↑)	0.9382	0.9608
AMI (↑)	0.8934	0.9419

SC: Silhouette coefficient, NMI: normalized mutual information, and AMI: adjusted mutual information.

V. Computational Complexity

We report the parameter count, training time, inference latency, and peak training GPU memory of the proposed method and AMAE in Table S3. The proposed method uses 4.24 M parameters, i.e., about 61% fewer than AMAE (10.93 M), at the cost of a longer training time (13.29 vs. 8.48 s/epoch), a slightly higher inference latency (0.41 vs. 0.33 ms/sample, about 27% higher), and a larger peak training GPU memory (10.35 vs. 4.90 GB), which mainly stems from the multi-layer aggregation and the attention-based decoder. In absolute terms the overhead is modest and is paid once at training time, and the inference latency in particular stays below one millisecond per sample. During inference, only the encoder and the classifier are retained.

TABLE S3: Computational overhead comparison.

Method	Params (M)	Training time (s/epoch)	Inference latency (ms/sample)	Peak GPU memory (MB)
Proposed	4.24	13.29	0.41	10349.66
AMAE	10.93	8.48	0.33	4898.24

VI. Paired Wilcoxon Signed-Rank Test Details

Since all methods are evaluated under the same 50 random seeds, the paired Wilcoxon signed-rank test is conducted on the improvements over the best-performing comparative method in each setting. The detailed results are listed in Table S4. The test is two-sided, paired over the same 50 seeds, and the p-values are Holm-corrected across the nine settings. The improvements are statistically significant ($p \leq 8.7\times10^{-3}$) in 8 of the 9 settings, the only exception being the ADS-B 10-shot setting, where the improvement over AMAE is not significant ($p = 0.25$), i.e., the case with the smallest margin and the largest relative variance, so no advantage is claimed there.

TABLE S4: Paired Wilcoxon signed-rank test results (50 shared seeds).

Setting	Best-performing baseline	Improvement	p (Holm)
ADS-B 10-shot	AMAE 66.05±3.54%	+1.07%	0.25
ADS-B 20-shot	AMAE 82.27±2.22%	+2.43%	3.0×10^{-3}
ADS-B 30-shot	AMAE 86.90±1.98%	+2.48%	5.0×10^{-5}
LoRa 10-shot	LoRaSONN 82.75±1.29%	+1.87%	8.7×10^{-3}
LoRa 20-shot	DIPA 89.25±1.57%	+3.14%	3.0×10^{-4}
LoRa 30-shot	DIPA 92.43±1.76%	+2.65%	1.1×10^{-3}
WiFi 10-shot	DIPA 85.61±1.64%	+4.81%	1.5×10^{-7}
WiFi 20-shot	DIPA 91.47±0.95%	+5.60%	7.2×10^{-9}
WiFi 30-shot	DIPA 96.14±0.44%	+2.29%	9.2×10^{-9}

VII. Gradient-Handling Ablation

We compare the stop-gradient design (adopted in the manuscript) with the EMA-teacher variant and the variant without stop-gradient on the full-view branch. The results are listed in Table S5. The adopted stop-gradient design performs on par with the "Stop-gradient + EMA teacher" variant, while removing the stop-gradient clearly degrades the identification accuracy (84.70±2.68% → 69.47±5.53% on ADS-B and 97.07±1.47% → 85.52±3.98% on WiFi), confirming the necessity of the stop-gradient operation.

TABLE S5: Gradient handling of the full-view branch.

Variant	ADS-B (SS)	WiFi (MS)
Stop-gradient (default)	84.70±2.68%	97.07±1.47%
Stop-gradient + EMA	85.42±2.51%	97.48±1.22%
w/o stop-gradient	69.47±5.53%	85.52±3.98%

TABLE S6: Downstream accuracy (%) when the encoder is pre-trained on the synthetic signals (10-seed mean±std) and on the real ADS-B auxiliary data (repeated from Table II).

	Real ADS-B		Synthetic	
Setting	AMAE	Proposed	AMAE	Proposed
ADS-B 10	66.05±3.54	67.12±3.92	58.54±5.24	65.96±2.62
ADS-B 20	82.27±2.22	84.70±2.68	71.34±3.05	81.40±3.54
ADS-B 30	86.90±1.98	89.38±1.39	86.04±1.92	89.14±1.45
LoRa 10	80.62±4.76	84.62±3.79	75.11±3.65	85.70±1.91
LoRa 20	89.25±1.57	92.39±3.45	83.38±2.29	90.73±3.91
LoRa 30	92.43±1.76	95.08±3.72	87.04±2.68	92.81±3.48
WiFi 10	85.61±1.64	90.43±3.06	81.27±2.88	89.90±4.81
WiFi 20	91.47±0.95	97.07±1.47	89.53±1.94	93.57±2.11
WiFi 30	96.14±0.44	98.43±0.66	93.35±0.51	97.72±0.73

VIII. Projection-Module Ablation (MLP vs. Linear)

We compare the learnable linear projection ϕ_l with a two-layer MLP projection in the hierarchical feature aggregation. The results show that the performance difference is negligible (84.35±2.75% vs. 84.70±2.68% under the ADS-B 20-shot setting), indicating that the gain mainly stems from the aggregation mechanism rather than the projection capacity.

IX. Pre-training on Synthetic Signals

The three public datasets do not provide a fourth signal type, so a controlled simulation is used to test whether the auxiliary data must be a real recording. A packet is generated by

$$s[n] = \textstyle\sum_k a_k \, g_\beta\big(\frac{n-kM}{M}\big)\,, \tag{5}$$

$$y_1[n] = \frac{s[n]}{\Big(1 + \big(|s[n]|/A\big)^{2p}\Big)^{1/(2p)}}, \tag{6}$$

$$y_2[n] = \alpha\, y_1[n] + \eta\, y_1^{*}[n], \tag{7}$$

$$y[n] = y_2[n]\, \mathrm{e}^{j2\pi\varepsilon 10^{-6} f_{\mathrm{rf}} n/f_{\mathrm{s}}} + c + w[n], \tag{8}$$

where a_k is drawn from the QPSK, 16QAM or 64QAM alphabet and g_β is the root-raised-cosine pulse with roll-off $\beta = 0.35$, $M = 10$ samples per symbol and $L = 4800$ samples at $f_{\mathrm{s}} = 10$ MS/s. In (6) a Rapp power amplifier is used with $p \in [2, 4]$ and $A = \sigma_s 10^{\delta/20}$, where σ_s is the signal RMS and $\delta \in [1, 3]$ dB the compression at σ_s. In (7) the IQ imbalance is modelled with $\alpha = \frac{1}{2}(1 + g\mathrm{e}^{j\phi})$, $\eta = \frac{1}{2}(1 - g\mathrm{e}^{j\phi})$, $g \in [0.5, 2]$ dB and $\phi \in [0.5, 3]^\circ$. In (8) $\varepsilon \in [-10, 10]$ ppm, c and $w[n]$ are the carrier offset, the local-oscillator leakage in $[-30, -20]$ dBc and white Gaussian noise at 10–25 dB, respectively. Each device draws $\{p, \delta, g, \phi, \varepsilon, c\}$ once and keeps them fixed for all of its packets, while the symbols, the initial phase and the noise are redrawn per packet. The set is min-max normalised as the real data and contains ten devices with 100 packets each, i.e., 1000 waveforms, one third of the 3000 used for the ADS-B pre-training. Only its unlabeled waveforms are used, for the pre-training stage.

Table S6 reports the downstream accuracy. With the synthetic auxiliary data, the proposed method stays within -3.5 to $+1.1$ points of the real ADS-B auxiliary data, whereas AMAE loses up to 10.9 points, and the proposed method outperforms AMAE in all nine settings by 3.1–10.6 points. The proposed pre-training therefore does not depend on the auxiliary data being a real recording of a specific signal type.